Chapter - 13 - Advanced Python 2

Virtual Environment:

An environment which is same as the system interpreter but is isolated from the other Python environments on the system.

Installation.

- To use virtual environments, we write:

Pip install virtualenv # Install the package.

- We create a new environment using

virtualenv myprojectenv # creates a new venv

- The next step after creating the virtual environment is to activate it.
  → .\<name>\Scripts\activate.ps1.
- → We can now use this virtual environment as a separate python installation.

Pip freeze Command:

'pip freeze' returns all the package installed in a given python environment along with the versions.

pip freeze > requirement.txt.

- The above Command creates a file named 'requirement.txt' in the same directory containing the output of 'pip freeze'
- We can distribute this file to other users, and they can recreate the same environ-ment using: pip install -r requirements.txt.

- **Lambda Functions :**
Functions created by using an expression using 'lambda' keyword.

Syntax :
Lambda arguments : expressions
can be used as a normal function.

Example :

```
Square = Lambda x : x * x
Square (6)   # returns 36
Sum = Lambda a, b, c : a + b + c
Sum (1, 2, 3)   # returns 6
```

- **Join Method (Strings)**
Creates a string from iterable objects.

```
L = ["apple", "mango", "banana"]
result = ", and, ".join (1)
print (result)
```

→ The above line will return "apple", and, "mango" and, banana".

Format Method (Strings)
Formats the values inside the strings into a desired output.

```
template . format (p1, p2...)
```

Syntax :

" {0} is a good {1}". format ("harry", "boy") # 1
" {1} is a good {0} format ("Micro", "boy") # 2

output 1
harry is a good boy

output 2
boy is a good Micro

Map, filter & reduce

Map applies a function to all the items in an input-list.

Syntax (•)

→ • map (function, input-List)
function can be lambda

→ Filter creates a list of items for which the function returns true.

• list (filter (function))

→ Reduce applies a rolling computation to sequential pair of elements.

• from functools import reduce
val = ~~replace~~ reduce (function, list)
# function can be lambda